

File permissions in Linux

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Project description

In this project, I worked as a security professional responsible for managing file and directory permissions within a Linux environment for a research organization. The goal was to review existing permissions, identify access control issues, and apply the appropriate authorization settings using Linux commands. By modifying file and directory permissions, I helped ensure that sensitive research data was protected and accessible only to authorized users.

Check file and directory details

```
researcher2@de59c954cf0d:~$ cd projects
researcher2@de59c954cf0d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 15:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 16:11 ..
-rw--w---- 1 researcher2 research_team  46 Jun  6 15:48 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  6 15:48 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun  6 15:48 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  6 15:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 15:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 15:48 project_t.txt
researcher2@de59c954cf0d:~/projects$
```

To check files and directory details I used `ls -la`. I added a `a` to the `ls-l` in order to see the hidden files and make sure every detail gets checked.

In the first 2 lines it prints the details of the directory and the upper directory. In the third line it displays the detail of a hidden file (a hidden file has a `.` before the filename). After the hidden files, the system displays the details of the directories inside the current directory, lastly, it shows the details of the files inside the directory.

Describe the permissions string

The permissions string is the first part of the details of a directory/file. It contains 10 characters:

The first character will always show if it's a directory or a file: If it's a directory it should be a `"d"`, if it's a file it will show a hyphen `"-"`.

The next 9 characters represent the permissions based on roles: From character 2-4 are the current user permissions, from 5-7 are the permissions for the group of the user and from 8-10 are the permissions for others.

There are 3 different permissions:

1. Read to read files, directories and subdirectories. (Represented by a “r”, if not conceded it will show a hyphen “-”).
2. Write to create files in directories and to edit files. (Represented by a “w”, if not conceded it will show a hyphen “-”).
3. Execute to launch programs and to enter directories. (Represented by a “x”, if not conceded it will show a hyphen “-”).

Example:

```
-rw--w---- 1 researcher2 research_team 46 Jun 6 15:48 .project_x.txt
```

.project_x.txt is a hidden file, not a directory. The user can read and write but can't execute, the group can only write while others can't do anything.

Change file permissions

As the organization does not allow others to have write access to any files, I need to remove the write permissions from others for project_k.txt

```
researcher2@cdf8e510be52:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 6 16:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 6 17:39 ..
-rw--w---- 1 researcher2 research_team 46 Jun 6 16:37 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 6 16:37 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Jun 6 16:37 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jun 6 16:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 6 16:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 6 16:37 project_t.txt
researcher2@cdf8e510be52:~/projects$ chmod o-w project_k.txt
researcher2@cdf8e510be52:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 6 16:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 6 17:39 ..
-rw--w---- 1 researcher2 research_team 46 Jun 6 16:37 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 6 16:37 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Jun 6 16:37 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jun 6 16:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 6 16:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 6 16:37 project_t.txt
researcher2@cdf8e510be52:~/projects$
```

For this I used `chmod o-w project_k.txt`, this command allows me to remove the permissions from others (o) using the minus (-) establishing which permission I wanted to remove write (w) and selecting the file `project_k.txt`.

Change file permissions on a hidden file

The research team has archived `.project_x.txt`, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. Use a Linux command to assign `.project_x.txt` the appropriate authorization.

```
researcher2@cdf8e510be52:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 16:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 17:39 ..
-rw--w---- 1 researcher2 research_team  46 Jun  6 16:37 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  6 16:37 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  6 16:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_t.txt
researcher2@cdf8e510be52:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@cdf8e510be52:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 16:37 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  6 17:39 ..
-r--r----- 1 researcher2 research_team  46 Jun  6 16:37 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  6 16:37 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  6 16:37 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  6 16:37 project_t.txt
researcher2@cdf8e510be52:~/projects$
```

For this I used `chmod u-w,g-w,g+r .project_x.txt`, this command allows me to remove the permissions from users (u) and group (g), using the minus (-) to establish that I wanted to remove write (w) permissions. It also allows me to add (+) the read permission to (g). When finished, it is necessary to write the file that we want to edit, in this case `.project_x.txt`.

Change directory permissions

The files and directories in the projects directory belong to the researcher2 user. Only researcher2 should be allowed to access the drafts directory and its contents.

```
researcher2@7f59c588358c:~$ cd projects
researcher2@7f59c588358c:~/projects$ ls -li
total 20
32297 drwx--x--- 2 researcher2 research_team 4096 Jun  6 19:41 drafts
32312 -rw-rw-rw- 1 researcher2 research_team  46 Jun  6 19:41 project_k.txt
32399 -rw-r----- 1 researcher2 research_team  46 Jun  6 19:41 project_m.txt
32609 -rw-rw-r-- 1 researcher2 research_team  46 Jun  6 19:41 project_r.txt
32611 -rw-rw-r-- 1 researcher2 research_team  46 Jun  6 19:41 project_t.txt
researcher2@7f59c588358c:~/projects$ chmod g-x drafts
researcher2@7f59c588358c:~/projects$ ls -li
total 20
32297 drwx----- 2 researcher2 research_team 4096 Jun  6 19:41 drafts
32312 -rw-rw-rw- 1 researcher2 research_team  46 Jun  6 19:41 project_k.txt
32399 -rw-r----- 1 researcher2 research_team  46 Jun  6 19:41 project_m.txt
32609 -rw-rw-r-- 1 researcher2 research_team  46 Jun  6 19:41 project_r.txt
32611 -rw-rw-r-- 1 researcher2 research_team  46 Jun  6 19:41 project_t.txt
researcher2@7f59c588358c:~/projects$
```

Here the groups had execution permissions for the drafts directory but only the user should access that so I used `chmod g-x drafts` in order to edit the permissions for the drafts directory.

Summary

In this activity, I examined file and directory details using Linux commands and analyzed the permissions assigned to users, groups, and others. I then modified permissions on files, hidden files, and directories using the `chmod` command to align with the organization's security requirements. These tasks helped ensure that sensitive project files and directories were accessible only to authorized users, protecting the research team's data and maintaining proper access control.